```
$: sudo cryptsetup luksFormat /dev/VolGroup/Root
```

Once you're done, you can unlock the volume with the following command:

```
$: sudo cryptsetup open --type luks /dev/VolGroup/Root
CryptRoot
```

This makes the encrypted volume available at */dev/mapper/CryptRoot*. Create a filesystem on it, as follows:

```
$: sudo mkfs.ext4 /dev/mapper/CryptRoot
```

Yes, remember to create a filesystem on the encrypted volume, not the LVM logical volume holding it. Now mount the volume with the command shown below:

```
$: sudo mount /dev/mapper/CryptRoot /mnt
```

That's it. Now what happens if you want to grow the logical volume? You begin by unmounting the filesystem and locking the volume, with the following command:

```
$: sudo umount /dev/mapper/CryptRoot
$: sudo cryptsetup luksClose /dev/VolGroup/Root
```

Then you resize the actual logical volume, with the following command:

```
$: sudo lvresize +10G VolGroup/Root
```

Then, unlock the encrypted volume again:

```
$: sudo cryptsetup open --type luks /dev/VolGroup/Root
CryptRoot
$: sudo umount /dev/mapper/CryptRoot # In case it was auto-
mounted
```

Resize the LUKS container:

```
$: sudo cryptsetup resize CryptRoot
```

Resize the filesystem within it:

```
$: sudo e2fsck -f /dev/mapper/CryptRoot
$: sudo resize2fs /dev/mapper/CryptRoot
```

And you're done. You can now go ahead and use that volume.

A detailed review of all the options available with LUKS is outside the scope of this article, and you'll have to refer to your distribution's manual on setting up an encrypted root using LVM and LUKS (or just ask around on StackOverflow, or better yet, Google around for a solution). But this should give you an idea of how to get started with encryption over LVM.

GRUB 2 and booting off LVM

If you're using GRUB 2, you can leave the */boot* partition inside LVM. Make sure you install GRUB to the MBR, and in your *grub.cfg* file, make sure the following are present:

1. Before the *menuentry* lines, you should have *insmod lvm*
2. Inside the *menuentry* section, specify your root like this: *set root=lvm/VolGroup-Root*

   Here's an example of a *grub.cfg* excerpt:

```
insmod lvm
…
menuentry "Arch Linux" {
     set root=lvm/VolGroup-Root
     linux /vmlinuz-linux root=/dev/VolGroup/Root rw quiet
     initrd /initramfs-linux.img
}
…
```

To conclude

There are three commands that you can use to get a quick overview of your LVM set-up. The first is *pvs*, and it displays information on your physical volumes. Shown below is the output of *sudo pvs* on my system:

```
  PV        VG       Fmt  Attr PSize   PFree
  /dev/sda3  VolGroup lvm2 a--  337.76g 157.76g
```

The next is *vgs*, and running it gives information about all the volume groups in the system. Here is its output on my system:

```
  VG    #PV #LV #SN Attr   VSize   VFree
  VolGroup   1   3   0 wz--n- 337.76g 157.76g
```

Finally, you have *lvs*, which shows you your logical volumes. This is what it says on my machine:

```
  LV   VG       Attr     LSize
  Arch VolGroup -wi-ao----  48.00g
  Home VolGroup -wi-ao---- 128.00g
  Swap VolGroup -wc-ao----   4.00g
```

You also have *pvdisplay*, *vgdisplay* and *lvdisplay* if you want detailed information.

That's all from me. Until next time, fellow Tuxers! END

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